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Atomic Gravimetry via Recursive Separation of Common- and Differential-Mode Interferometric Phases

Yida Wang, Liyuan Zhang, Hongzhou Xue, Tingting Lin, Yujuan Liu and Hongxu Ke

***Abstract*—Three-pulse Mach–Zehnder atom gravimeters typically employ alternating measurements with positive and negative effective wave vectors. By differencing the interferometric phases obtained for the two wave-vector directions, effective-wave-vector-independent phase components are suppressed and the gravity-related phase is extracted. However, conventional methods require the interferometric fringes corresponding to the positive and negative effective wave vectors to be fitted separately before phase differencing, which limits the update rate of the gravity estimates. To address this issue, this article proposes an atomic gravimetry method based on the recursive separation of common- and differential-mode interferometric phases. The differential-mode phase Φ_{dif}, which contains the gravity information, and the common-mode phase Φ_{com}, which represents the effective-wave-vector-independent phase component, are incorporated into a unified state-space model. An unscented Kalman filter (UKF) is then employed to recursively estimate these states from successive atomic population observations. A continuous 45-h static solid Earth tide monitoring experiment was conducted using an atom gravimeter, and the acquired data were processed offline to validate the proposed method. The experimental results demonstrate that the method can recursively separate the common- and differential-mode phases under a grouped alternating positive- and negative-effective-wave-vector measurement scheme. By updating the gravity-related state using each atom-drop observation, the gravity-estimate sequence obtained through offline processing has a result update interval of 0.6 s, compared with 24 s for the conventional least-squares (LS) method. Frequency-domain analysis shows that, within the low-to-mid-frequency noise plateau, the amplitude spectral density noise floor of the UKF gravity residuals is approximately $800\ \mu\mathrm{Gal}/\sqrt{\mathrm{Hz}}$, lower than the approximately $1500\ \mu\mathrm{Gal}/\sqrt{\mathrm{Hz}}$ obtained using the LS method. Allan deviation analysis further shows that the UKF results maintain a low deviation level over long averaging times, indicating good long-term stability.**

***Index Terms*—Atomic gravimetry; wave-vector-independent error; common- and differential-mode interferometric phase; unscented Kalman filter; least-squares fitting.**

I. Introduction

ATOM interferometric gravimetry has demonstrated tremendous application value in fields such as geophysical research, resource exploration, and aided inertial navigation, owing to its significant advantages of low drift operation, high precision, and continuous measurement capabilities [1]–[3]. It effectively circumvents the drawback of low measurement efficiency inherent in relative gravimeters, which require frequent calibration to address instrument drift [4], [5]. In recent years, the rapid development of engineering and miniaturization technologies for atomic gravimeters [6], [7] has gradually paved the way for their deployment in broader application scenarios, including vehicle-borne [8], [9] and shipborne measurements [10], [11].

Conventional atom gravimeters typically employ the least-squares (LS) method to separately fit the interferometric fringes corresponding to the positive and negative effective wave vectors, followed by differential processing to suppress the influence of effective-wave-vector-independent phase components [12], [13]. This fitting strategy, which uses complete interferometric fringe information, can achieve high gravity-estimation accuracy and offers clear advantages for station-based static gravity observations [14], [15]. However, because this method relies on the acquisition and fitting of complete interferometric fringes for both wave-vector directions, it limits the update rate of the gravity estimates while maintaining measurement accuracy. For applications that require the rapid acquisition of gravity information, the update rate of gravity estimation must be increased without compromising measurement accuracy [16], [17].

In practical applications, sensor measurements are generally affected by multisource noise and system nonlinearities [18], [19]. Recursive filtering algorithms based on optimal estimation theory have therefore been widely applied to nonlinear measurement systems [20]–[24]. In the field of atom interferometry, Kwolek *et al.* employed nonlinear Kalman filtering in a spatial-domain continuous atomic-beam interferometer with continuous readout and rapid wave-vector reversal, thereby separating the inertial phase from the wave-vector-independent phase component [25]. Recursive filtering methods have also been used to improve the efficiency of gravity estimation in atom gravimetry. Zhang *et al.* employed a square-root unscented Kalman filter to recursively estimate individual interferometric fringes and extracted gravity information from measurements with positive and negative effective wave vectors [26]. Jiang applied Kalman filtering to recursively estimate gravity measurement sequences obtained after conventional phase extraction and wave-vector-reversal processing [27], [28]. In a shipborne gravity measurement experiment, Dong *et al.* employed an extended Kalman filter to separately esti-

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mate the interferometric fringes for the positive and negative effective wave vectors and subsequently applied alternating averaging to the resulting gravity estimates [29]. Cheiney *et al.* used a Kalman filter to combine an atom interferometer with a classical accelerometer, thereby achieving vibration compensation and correcting the drift of the classical sensor [30]. However, for pulsed free-fall atom gravimeters operated with grouped alternating measurements using positive and negative effective wave vectors, existing recursive estimation methods still rely on the separate processing of the measurements from the two wave-vector directions or perform recursive estimation only after conventional phase extraction and wave-vector-reversal processing. Consequently, it remains difficult to directly update the gravity-related state using each atom-drop observation. Therefore, developing a recursive gravity-state estimation method based on individual atom-drop observations is important for increasing the update rate of gravity estimates from atom gravimeters.

For an atom gravimeter operated with grouped alternating measurements using positive and negative effective wave vectors, this article proposes a gravity measurement method based on the recursive separation of common- and differential-mode interferometric phases. The differential-mode phase Φ_{dif}, which contains the gravity information, and the common-mode phase Φ_{com}, which represents the effective-wave-vector-independent phase component, are incorporated into a unified state-space model. A nonlinear observation model is then established using the atomic population observations and the vibration compensation phase calculated from the vibration-sensor signal. An unscented Kalman filter (UKF) is employed to recursively estimate the state variables, thereby separating the common- and differential-mode phases. To account for possible differences in the interferometric fringe parameters between the two wave-vector directions, separate fringe amplitude and offset parameters are included in the state-space model for the positive and negative effective wave vectors. A continuous 45-h static solid Earth tide monitoring experiment was conducted using an atom gravimeter. The instrument was operated in a periodically repeated measurement sequence consisting of 20 consecutive positive-effective-wave-vector measurements followed by 20 consecutive negative-effective-wave-vector measurements. The acquired data were processed recursively offline to validate the common-mode/differential-mode phase separation and gravity-state estimation capabilities of the proposed method. Furthermore, the convergence performance of the proposed method under different initial conditions was analyzed, and the characteristics of the gravity residuals, the frequency-domain noise level, and the long-term stability were systematically evaluated.

II. METHODS

A. Observation Model of the Atomic Gravimetry System

In a three-pulse Mach–Zehnder atom gravimeter, cold-atom matter waves are manipulated through two-photon Raman transitions to realize the complete interferometric sequence of splitting, reflection, and recombination [31], [32]. By applying a linear frequency chirp to the Raman laser, the Doppler shift induced by the free fall of the cold-atom cloud is compensated, thereby maintaining effective Raman transitions throughout the interferometric process. The resulting interferometric phase contains information on gravitational acceleration, and the corresponding interferometric fringe can be expressed as

$$P = A + B\cos\left(\Phi_{\mathrm{obs}} - k_{\mathrm{eff}} g T^2\right) + v, \tag{1}$$

where P is the normalized atomic population; A and B denote the offset and amplitude of the interferometric fringe, respectively; k_{eff} is the effective wave vector; T is the time interval between adjacent Raman pulses; g is the gravitational acceleration; v is the detection noise; and Φ_{obs} is the observed phase, given by

$$\Phi_{\mathrm{obs}} = \Phi_{\mathrm{set}} - \Phi_{\mathrm{vib}}, \tag{2}$$

where $\Phi_{\mathrm{set}} = \alpha T^2$ is the preset scanning phase and α is the applied Raman frequency chirp rate. In this work, the positive and negative chirp scans correspond to measurements with the positive and negative effective wave vectors ($\pm k_{\mathrm{eff}}$), respectively. "Wave-vector reversal" refers to switching between the two effective-wave-vector directions, whereas "chirp scanning" refers to obtaining an interferometric fringe by varying the Raman frequency chirp rate for a given wave-vector direction. The vibration compensation phase Φ_{vib} is expressed as

$$\Phi_{\mathrm{vib}} = \int_{-\infty}^{+\infty} g(t)\frac{\mathrm{d}\phi_L(t)}{\mathrm{d}t}\,\mathrm{d}t \quad = k_{\mathrm{eff}}\int_0^{2T} g(t)v(t)\,\mathrm{d}t, \tag{3}$$

where $\phi_L(t)$ is the Raman laser phase, $v(t)$ is the vibration velocity of the Raman retroreflection mirror projected along the direction of the effective wave vector, and $g(t)$ is the phase sensitivity function of the atom interferometer [33].

For a single effective-wave-vector direction, the interferometric phase contains both the gravity-related phase and the effective-wave-vector-independent phase component. Conventional atom gravimetry generally employs effective-wave-vector reversal to obtain the interferometric phases Φ^+ and Φ^- corresponding to the positive and negative effective wave vectors, respectively. The gravity-related phase is then extracted by taking their half-difference:

$$\Phi_g = \frac{\Phi^+ - \Phi^-}{2}. \tag{4}$$

For measurements with positive and negative effective wave vectors, the gravity-related phase changes sign upon effective-wave-vector reversal, whereas the effective-wave-vector-independent phase component retains the same sign. Therefore, their half-difference, $\Phi_{\mathrm{dif}} = (\Phi^+ - \Phi^-)/2$, is defined as the differential-mode phase and represents the gravity-related phase component. Correspondingly, their half-sum, $\Phi_{\mathrm{com}} = (\Phi^+ + \Phi^-)/2$, is defined as the common-mode phase and represents the effective-wave-vector-independent phase component separated from the interferometric observations. Its variation may contain combined contributions from the single-photon light shift, second-order Zeeman shift, radio-frequency-related phase offsets, and other slowly varying wave-vector-independent terms.

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Based on the above phase decomposition, the nonlinear observation model of the system is expressed as

$$h(X, s) = a_s + b_s \cos\left(\Phi_{\text{obs}} - s\Phi_{\text{dif}} + \Phi_{\text{com}}\right), \tag{5}$$

where $s \in \{1, -1\}$ denotes the positive-effective-wave-vector ($s = 1$) and negative-effective-wave-vector ($s = -1$) measurement states. To separately describe the interferometric fringe characteristics for the two wave-vector directions, the fringe offset and amplitude parameters corresponding to the positive and negative effective wave vectors are independently included in the state vector as (a_p, b_p) and (a_n, b_n), respectively. The system state vector is defined as

$$X = \begin{bmatrix} a_p & b_p & a_n & b_n & \Phi_{\text{dif}} & \Phi_{\text{com}} & \dot{\Phi}_{\text{com}} \end{bmatrix}^{\text{T}}, \tag{6}$$

where $\dot{\Phi}_{\text{com}}$ denotes the rate of change of the common-mode phase Φ_{com}. To describe the slow low-frequency variation of the effective-wave-vector-independent phase component over time, the common-mode phase rate is included in the state vector, allowing the temporal evolution of Φ_{com} to be represented by the state-transition model.

Because the state parameters vary relatively slowly over a short sampling interval, the evolution of the state vector is described by a discrete-time linear stochastic process:

$$X_k = FX_{k-1} + w_k, \tag{7}$$

where X_k and X_{k-1} denote the system states at two consecutive sampling instants, with a physical time step of Δt; F is the state-transition matrix; and w_k is the process noise. A first-order slowly varying model is employed to describe the common-mode phase and its rate, and the stochastic process disturbances associated with the different state parameters are assumed to be statistically independent. The state-transition matrix F and process-noise covariance matrix Q are respectively expressed as

$$F = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & \Delta t \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \tag{8}$$

and

$$Q = \Delta t^2 \operatorname{diag}\left(\sigma_{a_p}^2, \sigma_{b_p}^2, \sigma_{a_n}^2, \sigma_{b_n}^2, \sigma_{\Phi_{\text{dif}}}^2, 0, \sigma_{\dot{\Phi}_{\text{com}}}^2\right), \tag{9}$$

where σ_{a_p}, σ_{b_p}, σ_{a_n}, σ_{b_n}, $\sigma_{\Phi_{\text{dif}}}$, and $\sigma_{\dot{\Phi}_{\text{com}}}$ denote the process-noise standard deviations of the corresponding state parameters. Their values are determined from the statistical characteristics of steady-state measurement data.

Based on the observation model $h(\cdot)$, the measurement equation at the kth sampling instant is expressed as

$$z_k = h(X_k, s_k) + v_k, \tag{10}$$

where z_k is the normalized atomic population observed at the kth sampling instant, v_k is the corresponding detection noise, and R is the measurement-noise variance. In this work, R is calibrated from the statistical characteristics of the steady-state interferometric-fringe-fitting residuals and is set to $R = 0.004$ in the experiment.

It should be noted that the state variables in (6) are not independently determined from a single scalar observation. Instead, they are recursively estimated from the time-series observations obtained through successive atom drops. As the chirp-scanning phase changes from one measurement to the next, the atomic population observations acquired at different sampling instants impose different constraints on the fringe parameters and phase states. Meanwhile, switching between the positive and negative effective wave vectors reverses the response sign of Φ_{dif} in the observation model, whereas the sign of Φ_{com} remains unchanged. The common-mode phase rate is further coupled to the temporal evolution of Φ_{com} through the state-transition model. Therefore, variations in the scanning phase, wave-vector polarity switching, and state evolution jointly provide multidimensional constraints over the successive observation sequence, allowing the different state parameters to be progressively identified and updated during recursive estimation.

B. Principle of the UKF

Equation (5) is a cosine-type nonlinear observation model. Applying the EKF would require the calculation of the Jacobian matrix and a first-order Taylor expansion, which may introduce linearization errors when the system nonlinearity is pronounced. Therefore, the UKF is adopted in this work. It employs the unscented transform (UT) to propagate a set of deterministic sigma points through the nonlinear observation function, thereby approximating the transformed mean and covariance without explicitly calculating the Jacobian matrix [34], [35]. The UKF estimation procedure consists of two successive steps: state prediction and measurement update.

1) State Prediction: Because the state-transition model is linear, the prior state and prior error covariance at the current time step are respectively expressed as

$$\hat{X}_{k|k-1} = F\hat{X}_{k-1|k-1}, \tag{11}$$

and

$$P_{k|k-1} = FP_{k-1|k-1}F^{\text{T}} + Q, \tag{12}$$

where $\hat{X}_{k-1|k-1}$ and $P_{k-1|k-1}$ denote the posterior state estimate and its error covariance matrix at the previous time step, respectively, whereas $\hat{X}_{k|k-1}$ and $P_{k|k-1}$ denote the prior state estimate and its error covariance matrix at the current time step, respectively.

2) Measurement Update: Based on the prior state mean $\hat{X}_{k|k-1}$ and covariance $P_{k|k-1}$, the scaled UT is used to construct $2L + 1$ sigma points $\chi_{i,k|k-1}$, where the state dimension is $L = 7$:

$$\chi_{0,k|k-1} = \hat{X}_{k|k-1}, \tag{13}$$

$$\chi_{i,k|k-1} = \hat{X}_{k|k-1} + \left[\sqrt{(L+\lambda)P_{k|k-1}}\right]_i, \quad i = 1, \ldots, L, \tag{14}$$

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and

$$\chi_{i,k|k-1} = \hat{X}_{k|k-1} - \left[\sqrt{(L+\lambda)P_{k|k-1}}\right]_{i-L}, \quad i = L+1,\ldots,2L, \tag{15}$$

where

$$\lambda = \alpha^2(L+\kappa) - L$$

is the scaling parameter. The parameter α controls the spread of the sigma points around the state mean and is set to $\alpha = 0.1$ in this work. The parameter β incorporates prior information on the higher-order moments of the state distribution. For a Gaussian distribution, $\beta = 2$ is generally adopted, and $\beta = 2.0$ is used in this work. The parameter κ is an auxiliary scaling parameter and is set to $\kappa = 0$.

The sigma points are subsequently propagated through the nonlinear observation model incorporating the current wave-vector polarity s_k:

$$\mathcal{Z}_{i,k|k-1} = h\left(\chi_{i,k|k-1}, s_k\right), \tag{16}$$

where $\mathcal{Z}_{i,k|k-1}$ denotes the predicted normalized atomic population corresponding to the ith sigma state point after propagation through the nonlinear observation model.

Using the corresponding mean weights $W_m^{(i)}$ and covariance weights $W_c^{(i)}$, the predicted measurement $\hat{z}_k$, measurement autocovariance P_{zz}, and state–measurement cross-covariance P_{xz} are respectively calculated as

$$\hat{z}_k = \sum_{i=0}^{2L} W_m^{(i)} \mathcal{Z}_{i,k|k-1}, \tag{17}$$

$$P_{zz} = \sum_{i=0}^{2L} W_c^{(i)} \left(\mathcal{Z}_{i,k|k-1} - \hat{z}_k\right)\left(\mathcal{Z}_{i,k|k-1} - \hat{z}_k\right)^{\mathrm{T}} + R, \tag{18}$$

and

$$P_{xz} = \sum_{i=0}^{2L} W_c^{(i)} \left(\chi_{i,k|k-1} - \hat{X}_{k|k-1}\right)\left(\mathcal{Z}_{i,k|k-1} - \hat{z}_k\right)^{\mathrm{T}}. \tag{19}$$

The Kalman gain is then calculated as

$$K_k = P_{xz} P_{zz}^{-1}, \tag{20}$$

and the system state and error covariance matrix are updated using the current atomic population observation z_k:

$$\hat{X}_{k|k} = \hat{X}_{k|k-1} + K_k\left(z_k - \hat{z}_k\right), \tag{21}$$

and

$$P_{k|k} = P_{k|k-1} - K_k P_{zz} K_k^{\mathrm{T}}. \tag{22}$$

Through the above state-prediction and measurement-update procedures, the UKF recursively updates the interferometric fringe parameters for the positive and negative effective wave vectors, the differential-mode phase, the common-mode phase, and its rate using successive atomic population observations. This enables continuous recursive estimation of the common- and differential-mode interferometric phases and updates the gravity-related state using each atom-drop observation.

III. Static Monitoring Experiment

Under steady-state laboratory conditions, a continuous 45-h static monitoring experiment was conducted using a free-fall $^{85}\mathrm{Rb}$ atom gravimeter. The experimental setup and signal-processing scheme are illustrated in Fig. 1. To balance the interferometric phase sensitivity and system dynamic range, the Raman interrogation time T was set to 41 ms, and the durations of the $\pi/2$ and π pulses were set to 7 μs and 14 μs, respectively. Each atom-drop measurement cycle lasted 0.6 s. The Raman chirp-rate scan step was 48.8 Hz/s, and the effective wave vector k_{eff} was approximately 1.61×10^7 rad/m. The typical number of trapped atoms was approximately 10^8, and the typical interferometric fringe contrast was approximately 17%.

A Nanometrics Titan TACCL-N1 triaxial force-balance accelerometer was used for vibration measurements in the experiment. It has a bandwidth from DC to 430 Hz (-3 dB) and a dynamic range of 166 dB at 1 Hz over a 1-Hz bandwidth. The vertical acceleration output was converted from voltage to acceleration using the manufacturer-provided sensitivity and transfer function and was sampled at 50 kHz by the data acquisition system. The vibration compensation phase Φ_{vib} was then calculated using the phase sensitivity function of the atom interferometer. Under the present static experimental conditions, the bandwidth and dynamic range of the accelerometer were sufficient for the vibration measurement requirements. Nevertheless, the residual vibration noise after compensation may still be affected by factors such as accelerometer self-noise and errors in vibration-phase reconstruction.

The Raman pulses and frequency control were synchronized using a unified hardware timing system, with a PRS10 rubidium frequency standard serving as the common time reference. During the experiment, grouped alternating measurements with positive and negative effective wave vectors were employed. Specifically, 20 consecutive measurements with the positive effective wave vector were followed by 20 consecutive measurements with the negative effective wave vector, and this sequence was repeated periodically. Each atom drop corresponded to one chirp-rate scan point. Since each atom-drop cycle lasted 0.6 s, each wave-vector measurement block lasted 12 s, and one complete positive/negative-effective-wave-vector measurement cycle lasted 24 s. The acquired experimental data were subsequently processed recursively offline using the proposed UKF method.

IV. Results and Discussion

A. Analysis of Vibration Noise and Vibration-Sensor Baseline

Fig. 2(a) shows the amplitude spectral density (ASD) of the ambient vibration acceleration derived from the vibration-sensor measurements during the monitoring period. It characterizes the external vibration disturbances affecting the installation environment of the experimental system and the support structure of the reference mirror. The spectral results indicate that, even under steady-state laboratory conditions, the reference mirror is inevitably affected by broadband vibration noise, with the ASD at the high-frequency resonance peaks reaching approximately 10^{-3} $(\mathrm{m/s^2})/\sqrt{\mathrm{Hz}}$. According to the

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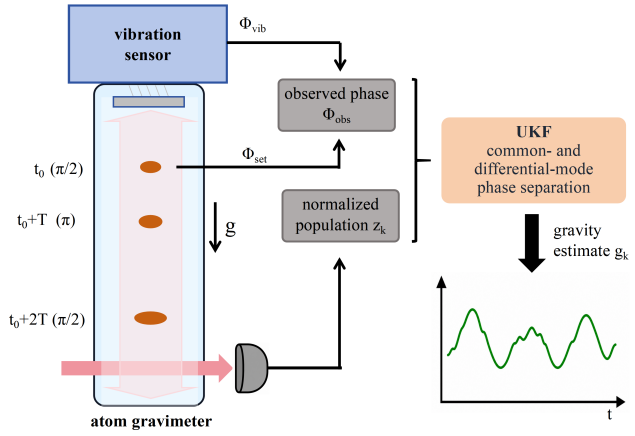


Fig. 1. Schematic of the atom-gravimeter measurement sequence and UKF-based recursive gravity-estimation framework. The vibration compensation phase Φ_{vib}, calculated from the signal measured by the broadband vibration sensor, is combined with the preset scanning phase Φ_{set} to obtain the observed phase Φ_{obs}. The normalized atomic population z_k serves as the measurement observation of the nonlinear state-space model. Based on these observations, the UKF recursively estimates the differential-mode phase Φ_{dif}, common-mode phase Φ_{com}, and interferometric fringe parameters, thereby updating the gravity estimate g_k.

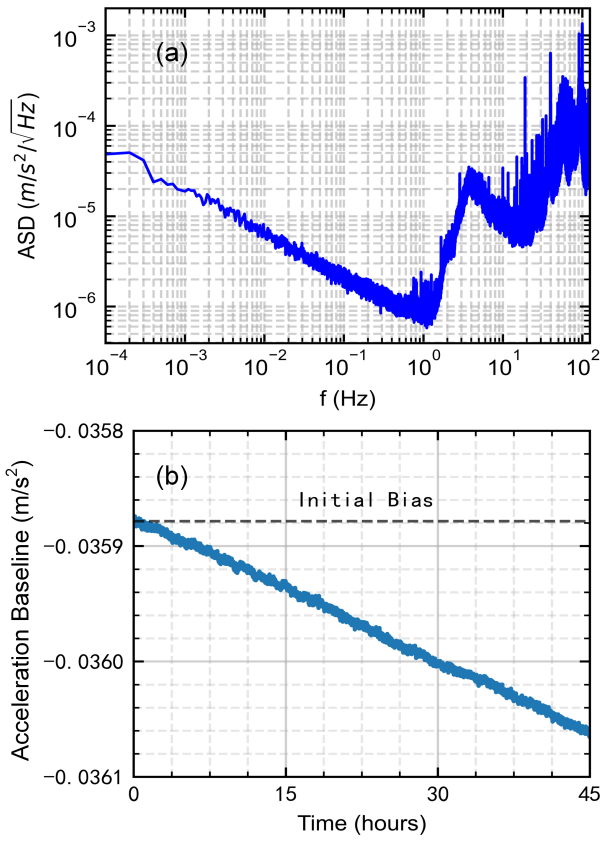


Fig. 2. Vibration characteristics during the static gravity measurement. (a) Amplitude spectral density of the measured acceleration signal. (b) Long-term variation of the acceleration baseline during the 45-h measurement.

mapping relationship between the reference-mirror vibration and the atom-interferometric phase, $\delta\Phi_{\mathrm{vib}} = k_{\mathrm{eff}}\delta z$, strong high-frequency vibration disturbances directly induce fluctuations in the interferometric phase. Vibration compensation is therefore required to reduce their influence on the gravity measurement results.Fig. 2(b) shows the low-frequency baseline drift of the acceleration signal during the continuous 45-h monitoring period. Under the influence of environmental disturbances and the release of mechanical stress in the supporting structure, the acceleration baseline exhibits a slowly varying drift, with a cumulative change of approximately 18 mGal relative to its initial zero-point offset. To address the low-frequency baseline drift in the vibration signal, moving-average detrending was employed to separate the low-frequency baseline component before calculating the vibration compensation phase. The acceleration signal after baseline-drift removal was then integrated with the phase sensitivity function $g(t)$ of the atom interferometer to calculate Φ_{vib}. It should be noted that moving-average detrending cannot guarantee the complete removal of all low-frequency residual components, and the remaining slowly varying components may affect the subsequent phase estimation.

B. Analysis of UKF Results and Statistical Characteristics

Fig. 3 presents the state-estimation trajectories of the UKF over the continuous 45-h observation period, comprising more than 250 000 sampling points. In Fig. 3(a), the interferometric fringe offsets and amplitudes corresponding to the positive and negative Raman wave vectors are estimated independently, and all state variables rapidly converge from their assigned initial values to steady states. The state-evolution trajectories show that the fringe amplitudes gradually decrease throughout the monitoring period, primarily because of the long-term attenuation of the Raman laser intensity. The abrupt increases in the amplitudes result from manual optimization of the fiber coupler during the monitoring period, which restored the Raman laser intensity. In addition, clear differences can be observed between the fringe offsets and amplitudes corresponding to the positive and negative wave vectors. This result supports the use of separate fringe parameters for the two wave-vector directions in the state-space model.

Figs. 3(b) and 3(c) show the recursive estimates of the differential- and common-mode interferometric phases, respectively. The two phases gradually converge from their preset initial values of 1 rad and 0 rad, respectively, to steady states. After convergence, the common-mode phase exhibits pronounced slowly varying low-frequency behavior, whereas the differential-mode phase remains relatively stable and represents the gravity-related information. It should be noted that the estimated common-mode phase represents the effective-wave-vector-independent phase component separated from the interferometric observations. Its evolution may reflect the combined contributions of multiple wave-vector-independent effects. In addition, slowly varying residual components remaining after vibration-signal preprocessing may also affect the common-mode state. Therefore, the common-mode phase is not uniquely attributed to any specific physical mechanism in this work. These results demonstrate that the established state-space model can recursively track the common- and

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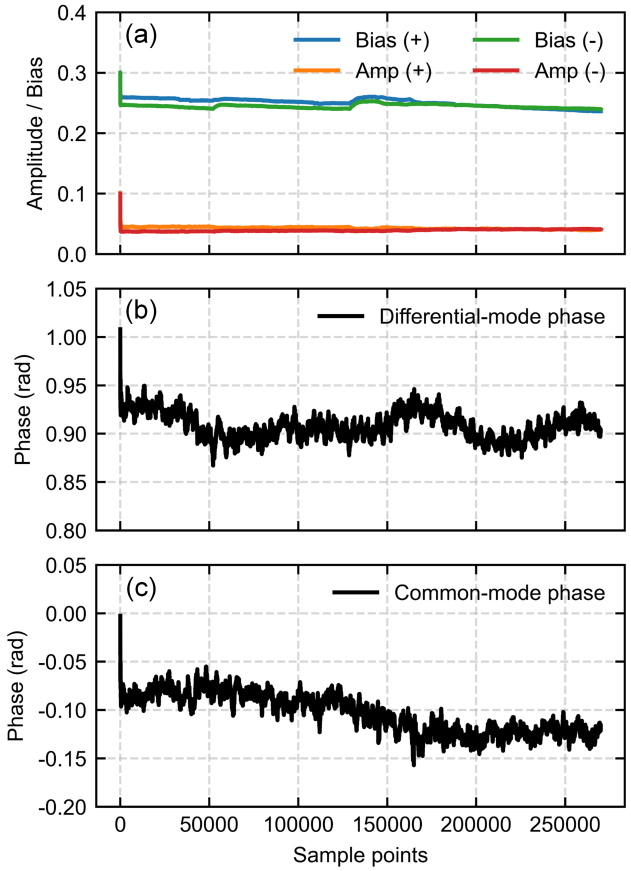


Fig. 3. UKF-estimated interferometric parameters and phase components during the 45-h measurement. (a) Estimated fringe amplitudes and offsets for the positive- and negative-effective-wave-vector measurements. (b) Differential-mode phase Φ_{dif}. (c) Common-mode phase Φ_{com}.

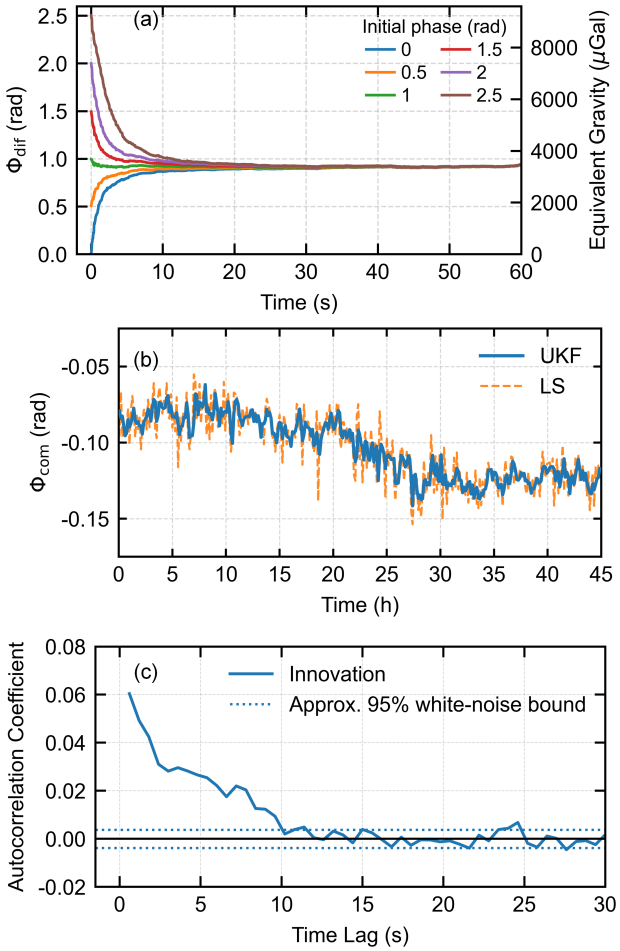


Fig. 4. Convergence and statistical characteristics of the UKF estimation. (a) Convergence of the differential-mode phase under different initial-phase conditions, with the corresponding equivalent gravity deviation shown on the right axis. (b) Comparison between the common-mode phase estimated by the UKF and that obtained using the LS method over the same time window. (c) Autocorrelation function of the UKF innovation sequence with 95% confidence bounds.

differential-mode phases during long-term continuous measurements.

To further evaluate the estimation performance of the proposed method, Fig. 4(a) shows the convergence of the differential-mode phase under different initial-phase conditions. The right vertical axis provides the equivalent gravity-deviation scale, for which a phase variation of 1 rad corresponds to approximately 3690 μGal. Different initial phases ranging from 0 to 2.5 rad were tested. After a transient period of approximately 20 s, all estimation trajectories smoothly converge to the same steady-state result, demonstrating that the proposed method is robust to the initial-phase setting. To validate the common-mode phase separation, the common-mode phase estimated by the UKF is compared with that constructed using the conventional LS method over the same time window, as shown in Fig. 4(b). The two results exhibit good agreement, with a Pearson correlation coefficient of 0.896, a mean difference of 0.000219 rad, and a root-mean-square error (RMSE) of 0.009756 rad. These results indicate that the UKF can effectively track variations in the effective-wave-vector-independent phase component contained in the interferometric observations. The UKF innovation sequence is further analyzed statistically, as shown in Fig. 4(c). Its autocorrelation decreases rapidly with increasing time lag and fluctuates approximately around zero after 10 s. The mean normalized innovation squared (NIS) is approximately 2.34, which is higher than the theoretical expectation of 1 for a scalar observation, indicating that the current model underestimates the actual observation-prediction errors to some extent. The measurement-noise covariance $R = 0.004$ was determined statistically from steady-state data and remained fixed throughout the measurement. However, the actual observation noise may vary with factors such as the number of atoms, fringe contrast, laser intensity, and residual vibration-compensation error. Therefore, the actual measurement noise during certain periods may be higher than the preset value. It should also be noted that, under the present grouped wave-vector-reversal scheme, the separation of the common- and differential-mode phases is constrained by the wave-vector-switching timescale. When the phase disturbances vary rapidly, coupling between the common- and differential-mode components may occur, and the effective separation bandwidth requires further evalu-

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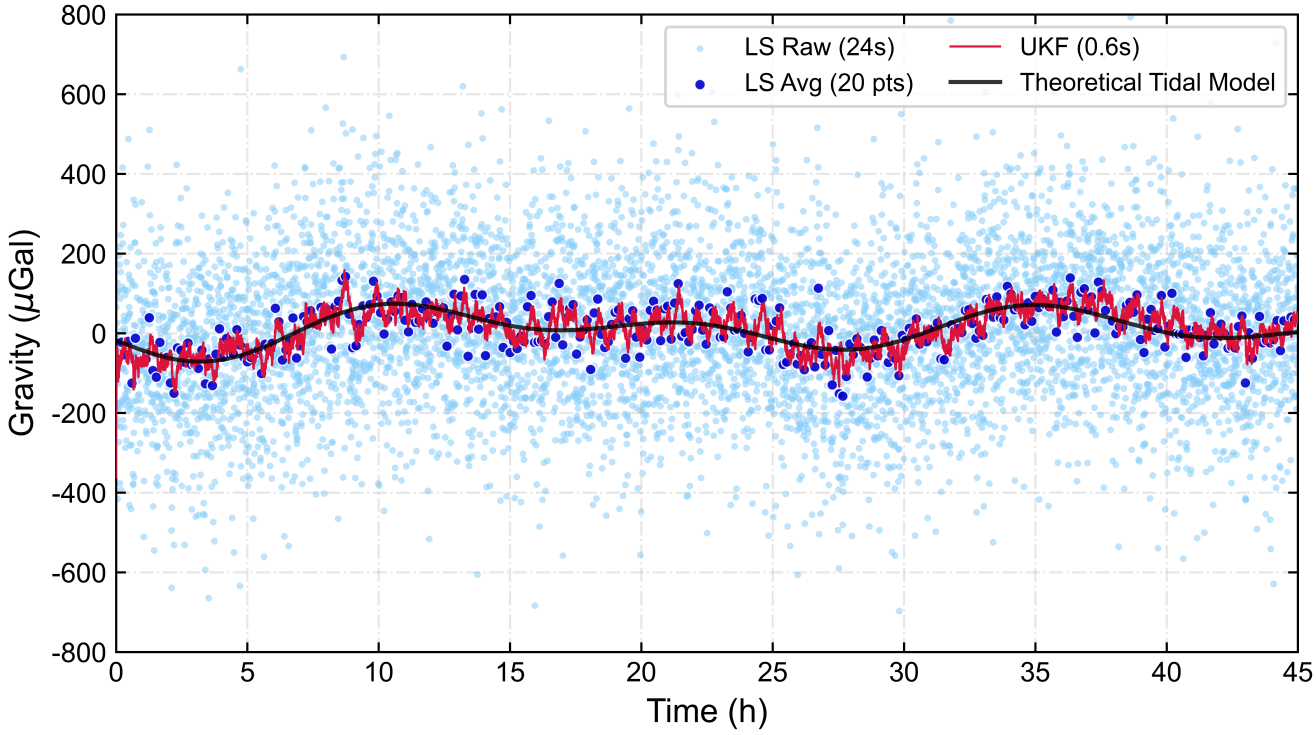


Fig. 5. Long-term gravity estimates during the 45-h static measurement. The UKF result with a 0.6-s update interval, the raw LS result with a 24-s update interval, and the averaged LS result are compared with the theoretical solid Earth tide used as the long-term trend reference.

ation.

C. Performance Evaluation of Gravity Measurement

Fig. 5 compares the gravity estimates obtained using the UKF and LS methods with the theoretical solid Earth tide curve over the continuous 45-h monitoring period. The theoretical solid Earth tide data were calculated using the Tsoft software based on the Wahr–Dehant model, together with the longitude, latitude, altitude, and Coordinated Universal Time (UTC) information of the monitoring site, and were used as a reference for the long-term gravity-variation trend. The conventional LS method requires the interferometric fringes for the positive and negative effective wave vectors to be fitted separately before phase differencing. Under the grouped alternating sequence employed in this work, which consists of 20 positive-effective-wave-vector measurements and 20 negative-effective-wave-vector measurements, each LS gravity estimate requires 24 s. In contrast, the UKF recursively updates the gravity-related state using successive atomic population observations, such that the gravity-estimate sequence obtained through offline processing has a result update interval of 0.6 s. LS Raw is obtained directly by fitting each complete group of interferometric fringes and exhibits pronounced point-to-point fluctuations. To obtain a more comparable time-averaged reference, LS Raw was further processed using 20-point block averaging to obtain LS Avg, corresponding to a 480-s time window. Its equivalent noise bandwidth is approximately 1.04 mHz, which is reasonably close to the empirical equivalent noise bandwidth of approximately 0.8 mHz for the UKF output, estimated by integrating the UKF residual power spectral density normalized to its low-frequency noise plateau. The results show that LS Avg can effectively reflect the solid Earth tide variation trend, although some residual fluctuations remain. The UKF result also agrees well with the theoretical solid Earth tide variation while providing a shorter result update interval.

As shown in Fig. 6, the ASD is evaluated for the gravity fluctuation data over the monitoring period. It should be noted that the gravity estimation data corresponding to the initial transient convergence phase of the UKF have been excluded from this spectral analysis. This frequency-domain analysis primarily reveals that within the specific solid Earth tide frequency band (around 10^{-5} Hz), the spectral density curves of the UKF and LS methods closely align, distinctly resolving the diurnal (O1/K1, approximately 1.16×10^{-5} Hz) and semidiurnal (M2, approximately 2.23×10^{-5} Hz) tidal components.

Table I summarizes the residual statistics of the different methods relative to the theoretical solid Earth tide model, with LS Avg serving as the time-averaged reference for comparison. The residual standard deviations (SDs) of LS Avg and UKF are 47.67 μGal and 32.28 μGal, respectively, indicating that the UKF output sequence exhibits a lower degree of residual dispersion. The corresponding RMSE values are 48.74 μGal and 34.05 μGal, respectively, demonstrating that the UKF result is in good agreement with the theoretical solid Earth tide variation trend. Furthermore, Fig. 7(a) presents the probability density distributions of the absolute residuals obtained using

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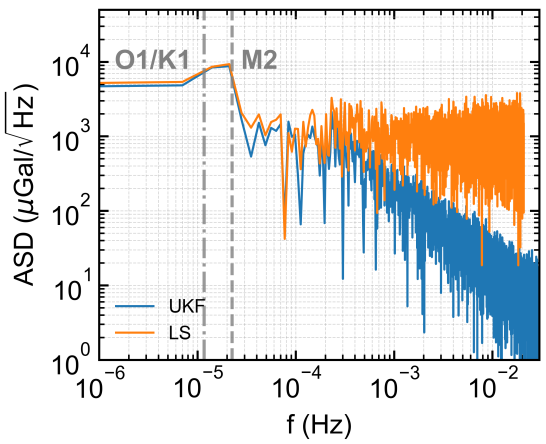


Fig. 6. Amplitude spectral density of the gravity estimates obtained using the UKF and LS methods. The spectral components associated with the principal solid Earth tide constituents O1/K1 and M2 are indicated for comparison.

the different methods. The mean absolute errors (MAEs) of LS Avg and UKF are 40.93 μGal and 24.29 μGal, respectively, with the UKF residuals exhibiting a more concentrated distribution. Overall, under reasonably comparable equivalent noise bandwidths, the UKF produces a lower gravity-residual level while maintaining a shorter result update interval.

The frequency-domain characteristics and temporal correlations of the gravity residuals are further analyzed. As shown in Fig. 7(b), the LS residuals exhibit an approximately flat ASD over the observed frequency band. Within the low-to-mid-frequency noise plateau, both the LS and UKF residuals exhibit an approximately f^0 noise-spectrum characteristic, as shown in Fig. 7(c). The noise floor of the LS residuals is approximately 1500 $\mu\mathrm{Gal}/\sqrt{\mathrm{Hz}}$, whereas that of the UKF residuals is approximately 800 $\mu\mathrm{Gal}/\sqrt{\mathrm{Hz}}$, indicating that the UKF residuals have a lower noise level within this frequency range. As the frequency increases, the ASD of the UKF residuals gradually rolls off, reflecting the frequency-response characteristics arising from recursive state propagation and the fusion of historical observations. Fig. 7(d) further presents the autocorrelation functions of the gravity residuals. The autocorrelation of the LS Raw residuals drops to near zero at the first nonzero lag (24 s) and subsequently fluctuates around zero, indicating no evident temporal correlation at nonzero lags. In contrast, the UKF residuals exhibit a slowly decaying positive autocorrelation over the displayed lag range, reflecting the temporal memory introduced by recursive state propagation and the fusion of historical observations. Therefore, the 0.6 s interval represents the update interval of the UKF gravity estimates rather than the interval between statistically independent estimates.

As shown in Fig. 8, the Allan deviations of the LS and UKF results are further compared with those obtained by applying Savitzky–Golay (SG; window length = 101 points) smoothing and Butterworth low-pass filtering ($f_c = 0.005$ Hz) to the LS gravity sequence. The SG and Butterworth methods represent two conventional post-processing approaches based on time-domain smoothing and frequency-domain low-pass filtering, respectively. The gray dashed line indicates the $\tau^{-1/2}$ reference slope, and the LS result approximately exhibits white-noise averaging behavior over short averaging times. After SG smoothing and Butterworth low-pass filtering, the Allan deviations exhibit an increasing trend over short-to-intermediate averaging times. This indicates that the smoothing and low-

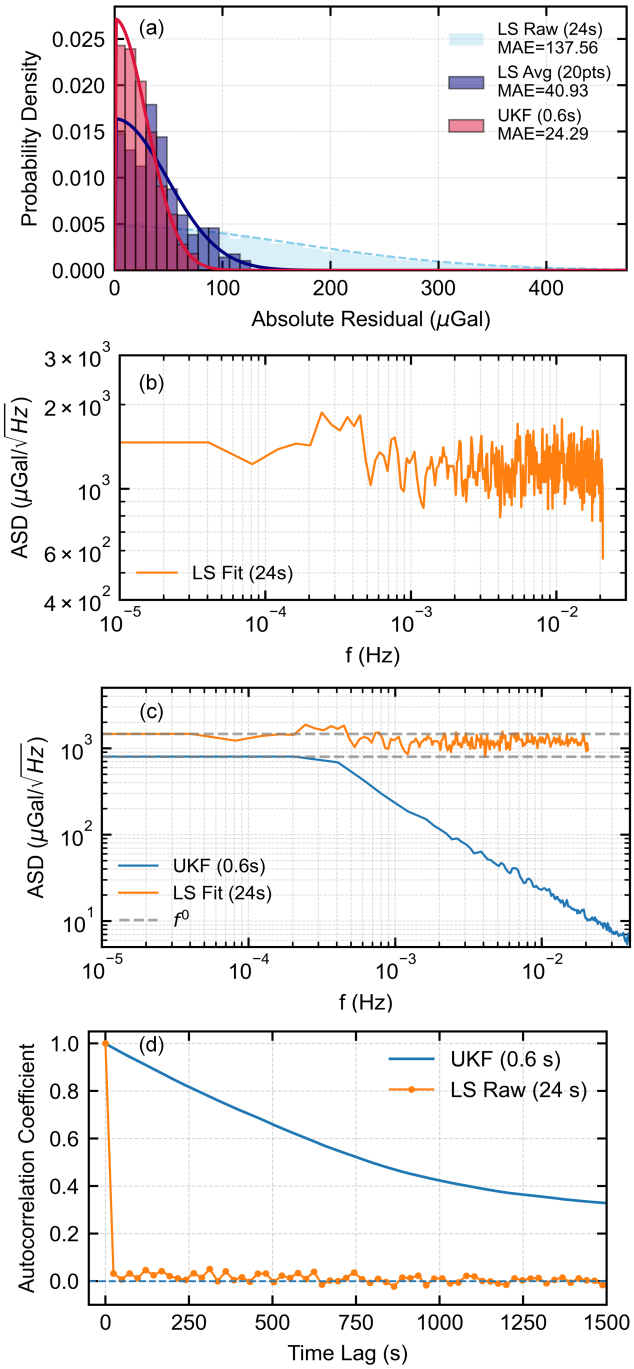


Fig. 7. Statistical, spectral, and temporal-correlation characteristics of the gravity residuals. (a) Probability density distributions of the absolute residuals obtained using LS Raw, LS Avg, and UKF. (b) Amplitude spectral density of the LS Raw gravity residuals. (c) Comparison of the residual amplitude spectral densities obtained using LS and UKF. (d) Autocorrelation functions of the LS Raw and UKF gravity residuals.

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TABLE I
COMPARISON OF RESIDUAL STATISTICS AMONG DIFFERENT METHODS.

Methods	RMSE (μGal)	SD (μGal)
LS Raw (24 s)	167.35	167.02
LS Avg (20 pts)	48.74	47.67
UKF (0.6 s)	34.05	32.28

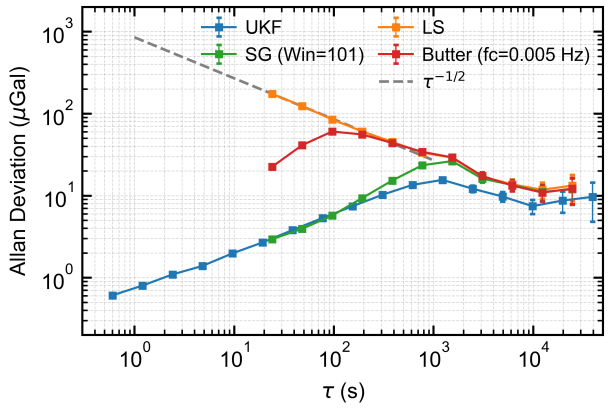


Fig. 8. Allan deviation of different gravity-estimation and post-processing methods. The Allan deviations of the UKF and LS results are compared with those obtained by applying Savitzky–Golay smoothing and Butterworth low-pass filtering to the LS gravity sequence. The dashed $\tau^{-1/2}$ line denotes the reference slope for white-noise averaging.

pass filtering operations alter the temporal-correlation structure of the original LS sequence and produce corresponding noise-reshaping effects. The UKF result exhibits similar non-white-noise behavior, with its Allan deviation increasing with averaging time over the short-to-intermediate range. This behavior is consistent with the long-term temporal correlation observed in Fig. 7(d) and primarily results from the reshaping of the output-noise characteristics by recursive state propagation and the fusion of historical observations. As the averaging time further increases, the Allan deviations of all methods gradually decrease, with the UKF result remaining generally low over long averaging times, indicating good long-term stability.

V. CONCLUSION

This article proposes and experimentally validates an atomic gravimetry method based on the recursive separation of common- and differential-mode interferometric phases. Under a grouped alternating measurement scheme with positive and negative effective wave vectors, the proposed method recursively estimates the common- and differential-mode phases from successive atomic population observations and updates the gravity-related state using each atom-drop observation. Consequently, the gravity-estimate sequence obtained through offline processing has a result update interval of 0.6 s. The effectiveness of the proposed method was validated through a continuous static solid Earth tide monitoring experiment. Compared with the LS method, the proposed method exhibits a lower ASD noise floor of the gravity residuals within the low-to-mid-frequency noise plateau and maintains a low Allan deviation level over long averaging times. These results provide a recursive data-processing approach for high-update-rate gravity-state estimation under grouped alternating measurements with positive and negative effective wave vectors and offer a reference for future investigations of online gravity estimation methods.

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